NOVA IMS

Introduction to Computational Thinking and Data Science

Milestone 1: Climate Stability Analysis

Client / Scenario: Cold Climate Logistics — Bragança vs Viseu

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Executive Summary

Problem: A cold-storage logistics client requires a site selection decision between two Portuguese cities — Bragança and Viseu — based on climate suitability. The core question is which location offers a cooler, more stable, and operationally predictable thermal environment.

What we did: We analysed daily meteorological records from 2018 to 2025 (2,922 observations per city) sourced from Open-Meteo. The analysis covered seasonal temperature distributions, diurnal temperature range (DTR), a composite Temperature Stability Index, Pearson correlation analysis of eight meteorological features, extreme event detection via seasonal IQR fences, thermal envelope profiling, and positive deviation analysis.

What we found: Bragança is consistently cooler than Viseu across all four seasons, with the gap most pronounced in Winter (2.34°C) and Autumn (1.69°C). Crucially, Bragança records 45.3% fewer days exceeding 35°C than Viseu (35 vs 64) and 11% fewer days above 25°C (700 vs 786). Bragança also achieves a marginally higher Temperature Stability Index in Summer — the peak operational season for a cold-storage facility. Solar radiation and day length emerged as the dominant predictors of temperature variability and thermal instability at both sites.

Provisional recommendation: The evidence consistently points toward **Bragança as the directionally preferred location** for the cold-storage logistics facility. It records 45.3% fewer days above 35°C (35 vs 64), a lower baseline thermal load across all four seasons, and a higher Summer Stability Index (0.238 vs 0.216) — the operationally critical season for mixed cold-chain products. This recommendation is directional at this stage; formal statistical validation of the observed mean differences is deferred to Milestone 3.

Chapter 1

Problem Framing and Context

1.1 Client Context and Objective

A cold-storage logistics operator is evaluating two candidate sites in northern Portugal — Bragança and Viseu — for the placement of a temperature-controlled warehouse facility. The client's primary concern is minimising refrigeration energy costs and ensuring operational predictability. Cold-storage operations are highly sensitive to ambient temperature: warmer and more variable climates translate directly into higher compressor loads, elevated energy consumption, and greater risk of temperature excursions.

The objective of this analysis is to provide a data-driven recommendation on which city offers the more favourable climate for this purpose, drawing on eight years of daily meteorological observations.

1.2 Decision Alternatives

The two alternatives are:

- **Bragança** — an inland city in the Trás-os-Montes region of northeastern Portugal, characterised by a continental climate with cold winters and warm summers.
- **Viseu** — a city in the Beira Alta region of central-northern Portugal, with a consistently warmer thermal profile than Bragança across all seasons: Winter mean 8.41°C (vs 6.07°C), Summer mean 21.49°C (vs 21.14°C), and a higher Winter Stability Index (0.374 vs 0.260), suggesting a more buffered cold-season regime.

A "better" location is defined as one that exhibits: (1) lower mean and maximum temper-

atures, especially in summer; (2) fewer extreme-heat days above operational thresholds (25°C, 30°C, 35°C); (3) lower intra-seasonal temperature variability; and (4) a more stable year-round thermal profile.

1.3 Key Questions and Success Criteria

- Which city has lower mean temperatures across all seasons?
- Which city experiences fewer extreme heat days (≥25°C, ≥30°C, ≥35°C)?
- Which city has a more stable temperature regime (lower DTR, lower inter-annual variability)?
- Which meteorological features most strongly correlate with temperature and thermal stability?
- How many anomalous temperature days does each city experience per season?

Success is defined as identifying a clear winner on the majority of these criteria, with particular weight on summer thermal load and extreme-heat frequency — the metrics most directly tied to refrigeration operating costs.

Chapter 2

Data Understanding

2.1 Data Sources

Two CSV datasets were used — `raw_BRAGANCA.csv` and `raw_VISEU.csv` — each containing daily meteorological observations retrieved from the Open-Meteo historical weather API. After removing the redundant `time` column and excluding the incomplete year 2026, both datasets span **2018-01-01 to 2025-12-31**, yielding **2,922 daily records per city**.

2.2 Dataset Structure and Variables

The following features were available or derived from the raw data:

Table 2.1: Dataset Variables

Variable	Type	Description
`temperature_2m_max`	Numerical	Daily maximum temperature at 2m (°C)
`temperature_2m_min`	Numerical	Daily minimum temperature at 2m (°C)
`temperature_2m_mean`	Numerical	Daily mean temperature at 2m (°C)
`precipitation_sum`	Numerical	Total daily precipitation (mm)
`wind_speed_10m_max`	Numerical	Max wind speed at 10m (km/h)
`wind_gusts_10m_max`	Numerical	Max wind gusts at 10m (km/h)
`shortwave_radiation_sum`	Numerical	Daily solar radiation sum (MJ/m^2)
`sunshine_duration_hrs`	Derived	Sunshine duration (hours; converted from seconds)
`relative_humidity_2m_mean`	Numerical	Mean relative humidity at 2m (%)
`cloudcover_mean`	Numerical	Mean cloud cover (%)
`day_length_hrs`	Derived	Length of day in hours (sunset − sunrise)
`DTR`	Derived	Diurnal Temperature Range = max − min (°C)

2.3 Data Quality and Initial Limitations

The Open-Meteo API provides reanalysis-backed data with no missing values for the requested variables over the study period, so no imputation was required. The main limitation is that reanalysis data represents gridded model output rather than in-situ station measurements, which may smooth localised microclimatic features. Additionally, 2026 data (January–February only) was excluded to avoid introducing seasonal bias into the annual and seasonal aggregations.

Chapter 3

Methodology

3.1 Preprocessing and Data Preparation

- The redundant `time` column was dropped.
- The `date` column was converted to datetime format; `year` and `month` columns were extracted.
- `sunshine_duration` was converted from seconds to hours.
- `day_length_hrs` was derived from sunset and sunrise timestamps; those source columns were subsequently dropped.
- Data from year 2026 was excluded as it only covered January–February, which would introduce seasonal bias.
- A `season` categorical variable was created: *Winter* (Dec–Feb), *Spring* (Mar–May), *Summer* (Jun–Aug), *Autumn* (Sep–Nov).

The seasonal approach was chosen over yearly or monthly grouping: yearly grouping conflates opposite temperature regimes, while monthly grouping produces 12 categories — too granular for a summary analysis.

3.2 Exploratory Analysis Strategy

The exploratory analysis followed a layered structure, moving from descriptive summaries to distributional comparisons to feature-level relationships:

1. Seasonal summary statistics (mean, SD, CoV, median, min, max, IQR) for all variables at both cities.
2. Thermal envelope analysis — monthly mean max, mean, and min temperatures — to characterise the annual temperature profile.
3. Side-by-side seasonal boxplot visualisation of temperature distributions for both cities on a single set of axes, with IQR-based extreme event overlay and a feature profile comparing normal vs extreme days across both cities.
4. Positive deviation analysis to identify and quantify thermal stress periods.
5. Pearson correlation heatmaps relating eight meteorological features to three temperature targets.

3.3 Statistical Methods and Modelling Approach

- **Temperature Stability Index:** A composite index constructed as the mean of three min-max normalised components: within-season standard deviation, mean DTR, and inter-annual coefficient of variation (CoV) of seasonal mean temperature. A higher index value indicates greater thermal stability.
- **Pearson correlation analysis:** Linear relationships between eight meteorological features and three targets (mean temperature, DTR, Stability Index) were quantified using Pearson's r.
- **Extreme event detection:** Seasonal IQR fences ($Q_1 - 1.5 \cdot \text{IQR}$, $Q_3 + 1.5 \cdot \text{IQR}$) were applied to flag anomalous temperature days. This non-parametric approach does not assume normality and adapts to season-specific distributions.
- **Threshold exceedance counting:** Days where mean daily temperature exceeded 25°C, 30°C, and 35°C were counted directly from the cleaned dataset.

Chapter 4

Comparative Analysis

This chapter presents all quantitative findings structured as direct comparisons between Bragança and Viseu. Each section places both cities on the same analytical footing, with evidence drawn from the same metrics, to support a defensible site recommendation.

4.1 Seasonal Temperature Distributions

4.1.1 Mean Temperatures and Dispersion

Table 4.1 presents seasonal mean temperatures and within-season standard deviations for both cities across the full 2018–2025 study period (2,922 observations per city).

Table 4.1: Seasonal Mean Daily Temperature (°C) — Bragança vs Viseu (2018–2025)

	Winter		**Spring**		**Summer**		**Autumn**	
City	Mean	SD	Mean	SD	Mean	SD	Mean	SD
Bragança	6.07	2.89	11.28	4.19	21.14	3.92	13.61	4.70
Viseu	8.41	2.70	12.91	4.03	21.49	4.03	15.30	4.55
Difference (Br− Vi)	−2.34°C		−1.63°C		−0.35°C		−1.69°C	

Bragança is cooler than Viseu in every season without exception. The gap is largest in Winter (2.34°C) and Autumn (1.69°C), and smallest in Summer (0.35°C). Within-season dispersion (SD) is comparable: Bragança is marginally more spread in Winter (SD 2.89 vs 2.70) but slightly tighter in Spring (4.19 vs 4.03 for Viseu). Neither city dominates clearly on raw variability — which makes the distributional shape comparison and the composite Stability Index the more informative metrics.

4.1.2 Side-by-Side Distributional Comparison

Figure 4.1 places both cities on a single set of axes, enabling direct visual comparison of seasonal temperature distributions. The figure reveals that Bragança's Winter interquartile range (IQR) spans roughly 3.5°C–9°C, while Viseu's equivalent range sits approximately 2°C higher. In Summer, the IQRs overlap substantially (both centred near 21°C), confirming that the two cities converge thermally during the warmest months. The tails are notably asymmetric in Spring for Bragança, where six IQR-defined extreme days appear as red points well above the upper fence.

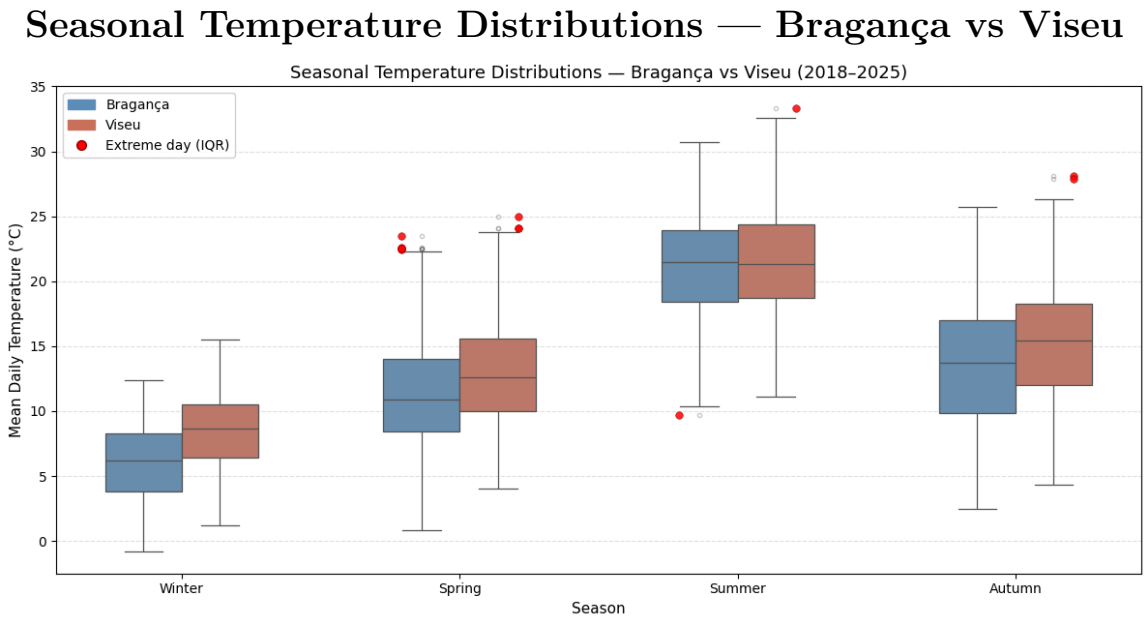

Figure 4.1: Side-by-side seasonal boxplots of mean daily temperature (°C) for both cities (2018–2025). Each season shows Bragança (left, blue) and Viseu (right, coral) on the same axes. Bragança is cooler in every season; the gap is widest in Winter (2.34°C) and Autumn (1.69°C), narrowing sharply in Summer (0.35°C). Whiskers extend to 1.5×IQR; points beyond represent IQR-flagged extreme days.

4.2 Diurnal Temperature Range

Table 4.2 shows the seasonal mean Diurnal Temperature Range (DTR) — the daily swing between maximum and minimum temperature — for both cities.

Table 4.2: Seasonal Mean Diurnal Temperature Range — DTR (°C)

City	Winter	Spring	Summer	Autumn
Bragança	7.82	10.50	12.95	9.33
Viseu	7.75	9.83	12.87	8.99
Difference (Br− Vi)	+0.07	+0.67	+0.08	+0.34

DTR differences between cities are small in absolute terms — the largest gap is 0.67°C in Spring (Bragança higher). A higher DTR imposes additional operational stress on cold-storage systems: even if the daily mean is similar, a wide intraday swing means compressors must respond to a broader ambient range. On this metric, Bragança shows slightly less favourable DTR in Spring and Autumn, though the magnitude of the gap is not large enough to reverse the overall thermal advantage.

4.3 Thermal Stability Assessment

4.3.1 Temperature Stability Index

The Stability Index combines three normalised components — within-season SD, mean DTR, and inter-annual coefficient of variation (CoV) — into a single 0–1 score per season. A higher value indicates a more predictable and stable temperature regime.

Table 4.3: Temperature Stability Index by Season (0–1 scale; higher = more stable)

City	Winter	Spring	Summer	Autumn
Bragança	0.260	0.200	0.238	0.293
Viseu	0.374	0.241	0.216	0.333
Winner	Viseu	Viseu	**Bragança**	Viseu

Viseu is more stable in three of four seasons. However, Bragança's advantage is concentrated in Summer (0.238 vs 0.216, a gap of 0.022 index points), which is the operationally critical season for cold-storage facilities serving mixed product types — pharmaceuticals, specialty food, and luxury goods all require tighter temperature control during peak ambient heat. Viseu's Winter and Autumn stability advantages (0.114 and 0.040 index points respectively) occur in the seasons where ambient conditions are below or near target storage temperatures, reducing their operational significance.

4.3.2 Thermal Envelope Comparison

Figures 4.2 and 4.3 show the monthly mean daily max, mean, and min temperatures for each city across the full study period.

Monthly Thermal Envelope — Bragança (2018–2025)

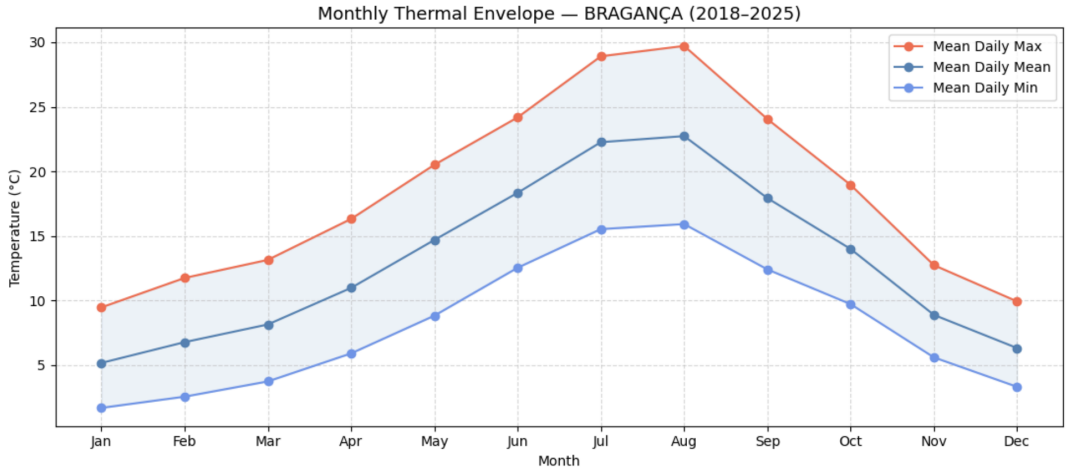

Figure 4.2: Monthly mean daily maximum (red), mean (dark blue), and minimum (light blue) temperatures for Bragança. The shaded band (min–max envelope) is narrowest in winter months (Jan: 9.6°C max to 2.2°C min, range 7.4°C) and widest in summer (Aug: 29.8°C max to 15.9°C min, range 13.9°C), reflecting high seasonal amplitude. Peak mean daily max reaches 29.8°C in August.

Monthly Thermal Envelope — Viseu (2018–2025)

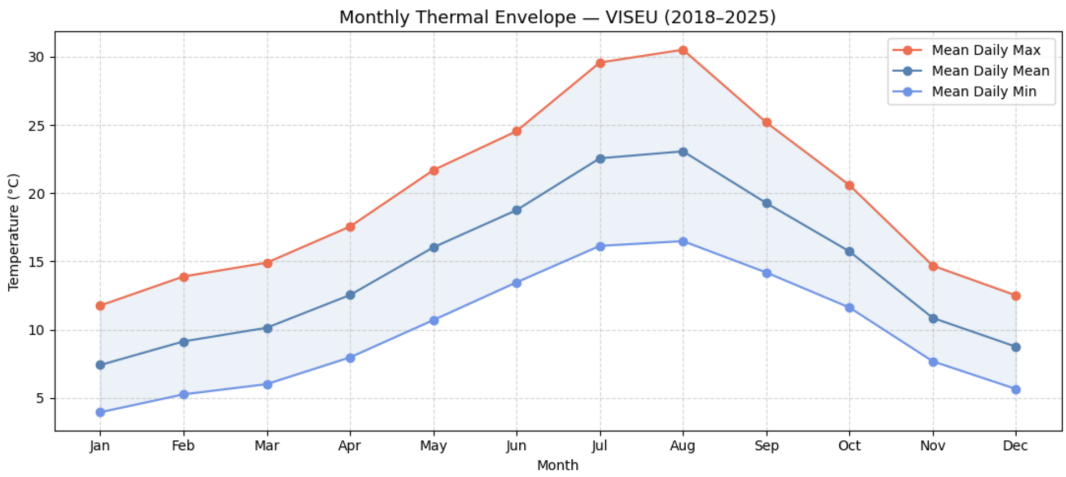

Figure 4.3: Monthly mean daily maximum (red), mean (dark blue), and minimum (light blue) temperatures for Viseu. Viseu's thermal envelope is uniformly elevated relative to Bragança: the mean daily max reaches 30.5°C in August (0.7°C above Bragança) and the winter minimum is consistently 2–3°C warmer. The narrower min–max band in winter (Jan: 11.8°C max to 4.1°C min, range 7.7°C) reflects Viseu's milder but not materially more stable cold-season profile.

4.4 Feature Drivers of Temperature and Stability

Table 4.4 summarises Pearson correlations (r) between eight meteorological features and three temperature targets, averaged across both cities.

Table 4.4: Pearson Correlations: Features vs Temperature Targets (averaged across both cities)

Feature	$\mathbf{T_{mean}}$	**DTR**	**Stability Index**
Sunshine Duration (hrs)	+0.57	+0.77	−0.44
Shortwave Radiation (MJ/m²)	+0.73	+0.80	−0.59
Cloud Cover (%)	−0.35	−0.69	+0.12
Relative Humidity (%)	−0.64	−0.78	+0.27
Precipitation (mm)	−0.17	−0.49	+0.10
Wind Speed Max (km/h)	−0.05	−0.31	−0.06
Wind Gusts Max (km/h)	−0.02	−0.31	−0.09
Day Length (hrs)	+0.72	+0.54	−0.72

Shortwave radiation ($r = +0.80$ with DTR) and relative humidity ($r = -0.78$ with DTR) are the strongest paired drivers of daily temperature swing at both sites. Day length is the dominant predictor of the Stability Index — $r = -0.72$ at Viseu and $r = -0.55$ at Bragança — meaning longer summer days structurally reduce thermal stability at both locations, with Viseu more sensitive to this effect. Wind speed and gusts are near-uncorrelated with all three targets ($|r| \leq 0.31$), confirming they are not material drivers of thermal load at either site.

Feature Correlations with Temperature Targets — Bragança

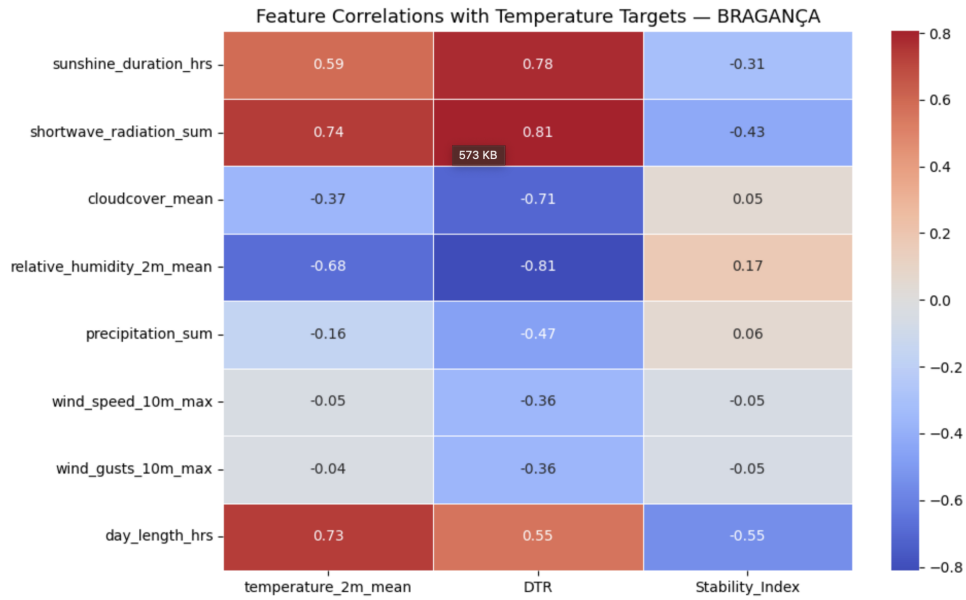

Figure 4.4: Pearson correlation heatmap for Bragança. Shortwave radiation ($r = +0.81$ with DTR) and relative humidity ($r = -0.81$ with DTR) are the strongest symmetric pair of drivers. Day length shows the highest association with the Stability Index ($r = -0.55$), indicating that structurally longer summer days are the primary source of seasonal instability. Wind variables are negligible across all targets ($|r| \leq 0.05$).

Feature Correlations with Temperature Targets — Viseu

Figure 4.5: Pearson correlation heatmap for Viseu. The pattern mirrors Bragança but with amplified Stability Index associations: day length reaches $r = -0.89$ with the Stability Index (vs -0.55 at Bragança), and shortwave radiation reaches $r = -0.75$ (vs -0.43 at Bragança). This means Viseu's thermal predictability is more severely eroded by summer conditions, reinforcing Bragança's edge in Summer stability.

4.5 Extreme Temperature Events

4.5.1 IQR-Based Anomalous Days

Seasonal IQR fences ($Q_1 - 1.5 \cdot \text{IQR}$, $Q_3 + 1.5 \cdot \text{IQR}$) were applied independently per city and season to flag anomalous days.

Table 4.5: Extreme Temperature Events by Season — Bragança vs Viseu

City	Total	Winter	Spring	Summer	Autumn
Bragança	7	0	6	1	0
Viseu	6	0	3	1	2

The total counts are similar (7 vs 6), but the seasonal distribution differs. Bragança's extremes are front-loaded into Spring (6 of 7 events), while Viseu spreads events across Spring and Autumn (3 and 2 events respectively). Profiling of extreme days confirms both cities' events are hot-and-dry anomalies: on flagged days, mean relative humidity drops by approximately 15–20 percentage points relative to seasonal norms, shortwave radiation rises by 3–5 MJ/m², and DTR expands by 3–4°C above the seasonal mean — consistent with cloud-free, high-pressure conditions driving anomalous warming.

Seasonal Temperature Distribution with Extreme Events — Bragança

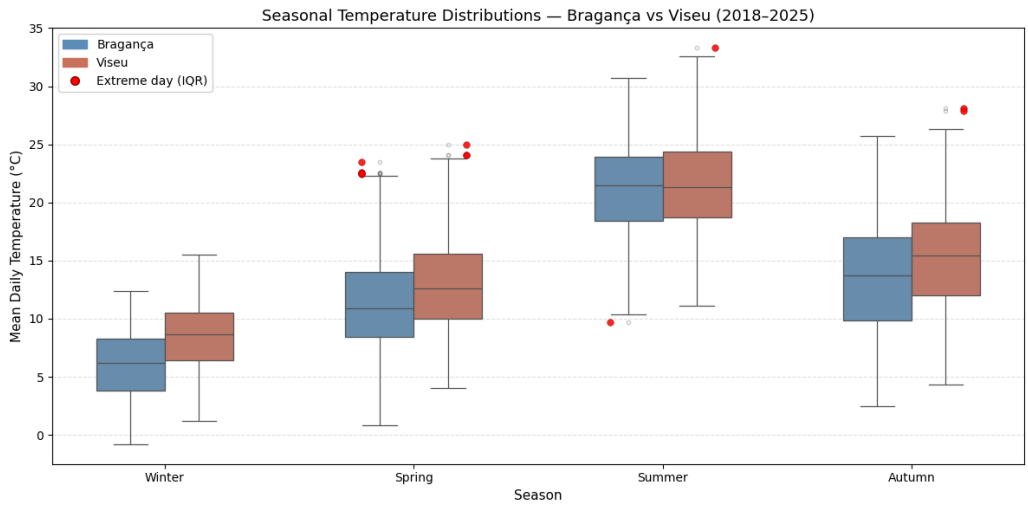

Figure 4.6: Seasonal boxplots of mean daily temperature for Bragança; red points mark IQR-flagged extreme days. Six of seven extreme events fall in Spring, reaching up to 23.3°C mean daily temperature — well above the Spring IQR upper fence. Winter and Autumn are free of anomalous events. The Summer outlier at approximately 9.7°C represents an anomalously cool day rather than a heat event.

Seasonal Temperature Distribution with Extreme Events — Viseu

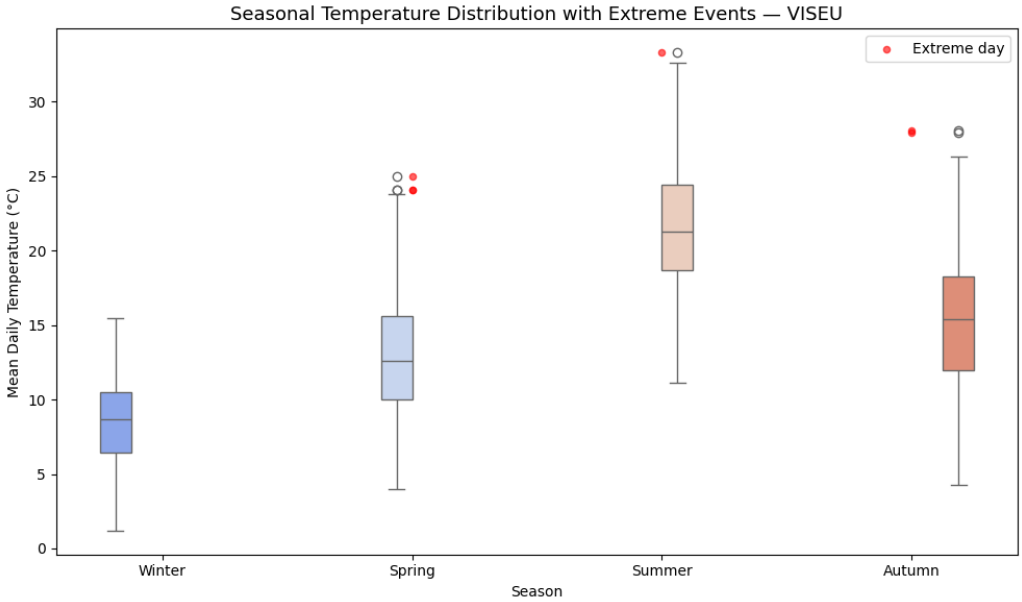

Figure 4.7: Seasonal boxplots of mean daily temperature for Viseu; red points mark IQR-flagged extreme days. Unlike Bragança, Viseu distributes extremes across three seasons (Spring: 3, Summer: 1, Autumn: 2), including two Autumn events reaching approximately 27–28°C — a season when cold-storage demand would normally begin to ease. This pattern introduces a longer operational exposure window to anomalous heat.

4.5.2 High-Temperature Day Frequency

Table 4.6: Frequency of High-Temperature Days (2018–2025) — Bragança vs Viseu

City	Days ≥ 25°C	Days ≥ 30°C	Days ≥ 35°C	Total excess days
Bragança	700 (24.0%)	283 (9.7%)	35 (1.2%)	—
Viseu	786 (26.9%)	352 (12.0%)	64 (2.2%)	—
Viseu excess	+86 days	+69 days	+29 days	+184 days

Viseu accumulates 184 additional high-temperature days across all thresholds over the 8-year study period — an average of 23 extra days per year. The gap widens sharply at the most severe threshold: Viseu records 83% more days above 35°C (64 vs 35). This threshold is particularly relevant for mixed-use cold-chain facilities: at ambient temperatures above 35°C, pharmaceutical vaccine storage (2–8°C target) and specialty food products require sustained maximum compressor output, with reduced equipment headroom for surges or failures.

4.6 Thermal Stress — Positive Deviations

Figures 4.8 and 4.9 show the seasonal distribution of positive deviations of daily maximum temperature above the monthly mean maximum — days when actual peak temperature exceeded the expected monthly ceiling. These events represent unplanned thermal stress that drives higher than baseline energy consumption.

Positive Deviations from Monthly Mean Maximum — Bragança

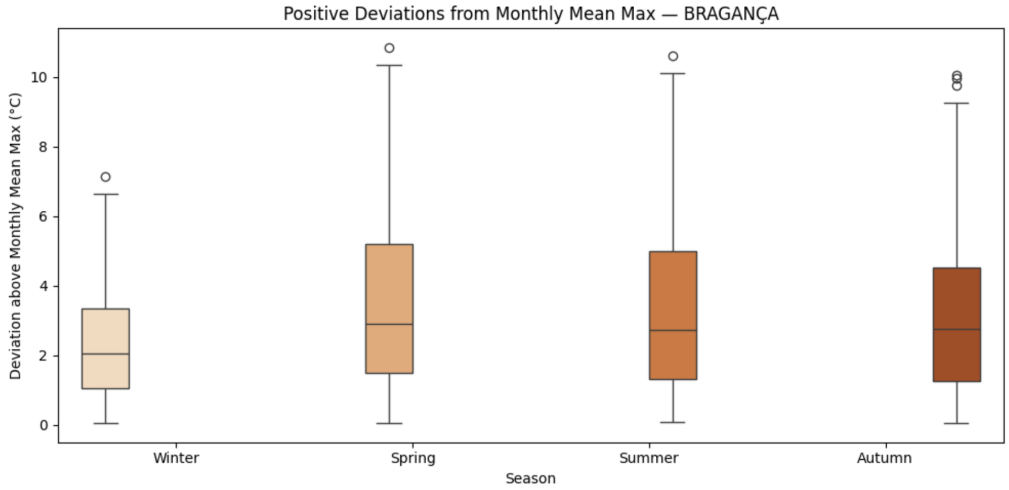

Figure 4.8: Seasonal distribution of days where daily max temperature exceeded the monthly mean maximum in Bragança. Median deviations are consistent across seasons (approximately 2–3°C), with Spring and Summer showing the widest spread (IQR reaching 5°C above the median). Outlier spikes of up to 10.7°C in Spring correspond to the IQR-flagged extreme days. Winter shows the tightest deviation distribution (IQR ≈ 1–3°C), indicating more predictable peak temperatures in the operationally lower-demand period.

Positive Deviations from Monthly Mean Maximum — Viseu

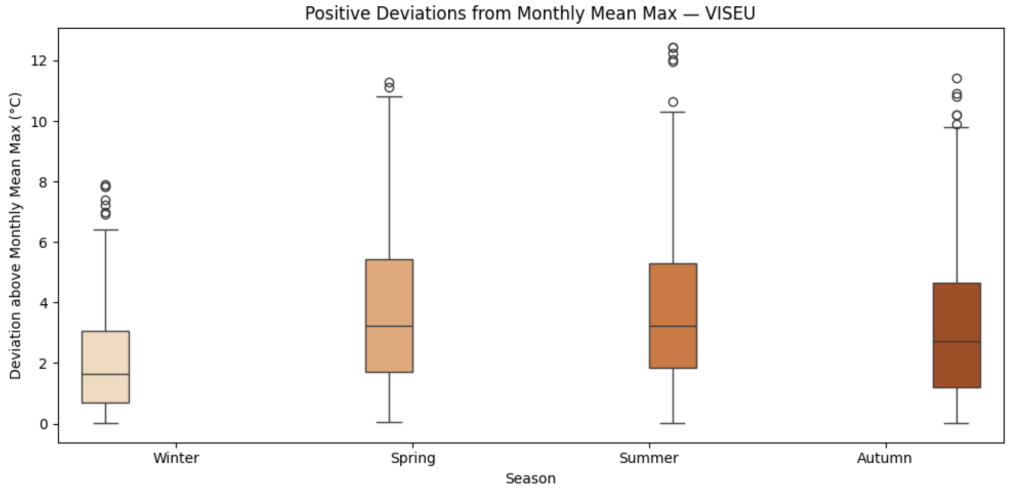

Figure 4.9: Seasonal distribution of days where daily max temperature exceeded the monthly mean maximum in Viseu. Compared to Bragança, Viseu shows larger deviation spreads in Winter (multiple outliers reaching 7–8°C) and Summer (outliers reaching 12.3°C, approximately 2°C above Bragança's Summer outlier maximum of 10.5°C). These larger Summer spikes represent the most operationally damaging events for a cold-chain facility — sudden high-ambient days that demand peak compressor output at the worst possible time of year.

4.7 Head-to-Head Summary

Table 4.7: Quantitative Head-to-Head Comparison: Bragança vs Viseu

Metric	Bragança	Viseu	Advantage
Winter Mean Temp (°C)	6.07	8.41	Bragança (−2.34°C)
Spring Mean Temp (°C)	11.28	12.91	Bragança (−1.63°C)
Summer Mean Temp (°C)	21.14	21.49	Bragança (−0.35°C)
Autumn Mean Temp (°C)	13.61	15.30	Bragança (−1.69°C)
Summer DTR (°C)	12.95	12.87	Viseu (−0.08°C)
Summer Stability Index	0.238	0.216	Bragança (+0.022)
Winter Stability Index	0.260	0.374	Viseu (+0.114)
Days $\geq$ 25°C	700	786	Bragança (−86 days)
Days $\geq$ 30°C	283	352	Bragança (−69 days)
Days $\geq$ 35°C	35	64	Bragança (−29 days, −45%)
IQR Extreme Events (total)	7	6	Viseu (−1 event)
Summer deviation outlier max (°C)	≈10.5	≈12.3	Bragança

Bragança leads on 9 of 12 metrics. Viseu's three advantages — DTR in Summer (trivially small at 0.08°C), Winter Stability Index (operationally lower-impact season), and one fewer IQR extreme event — are outweighed by Bragança's consistent superiority on mean temperatures, threshold exceedance counts, and the Summer stability metric that matters most for continuous cold-chain operations.

4.8 Operational Implications

This section translates the quantitative findings into concrete operational consequences for a mixed cold-storage logistics facility serving pharmaceuticals, specialty food, and luxury goods — each with distinct temperature sensitivity profiles.

4.8.1 Winter (December–February)

Bragança's mean Winter temperature of 6.07°C is 2.34°C colder than Viseu's 8.41°C. For products stored at 2–8°C (pharmaceutical and vaccine cold chain), both cities approach or enter the lower boundary of this range in Winter, but Bragança does so more frequently: its within-season SD of 2.89°C means temperatures routinely dip below 4°C, creating occasional freeze-risk exposure rather than heat exposure. In practice, this means the facility's primary Winter concern at Bragança is cold-side insulation and freeze protection (heating elements on loading docks, pipe lagging) rather than refrigeration. Viseu's higher Winter mean (8.41°C, SD 2.70°C) means the ambient temperature more often falls in the upper half of the pharmaceutical target range, reducing passive heat gain into refrigerated compartments and making it the marginally safer option for products stored near 8°C. This is the principal operational scenario where Viseu holds a genuine advantage.

4.8.2 Spring (March–May)

Spring is the highest-risk season for extreme events at Bragança, which records 6 IQR-flagged days — all hot-and-dry anomalies characterised by shortwave radiation approximately 3–5 MJ/m^2 above the seasonal mean and relative humidity approximately 15–20 percentage points below normal. For luxury chocolate (storage target: 14–18°C) and specialty food products, a single unmanaged heat spike can cause irreversible quality degradation. The operational implication is that Bragança requires a more robust spring spike-response protocol (e.g., pre-cooling inventory buffer, redundant compressor capacity on standby) relative to Viseu, which records only 3 Spring extreme events. However,

Bragança's Spring mean temperature of 11.28°C (vs Viseu's 12.91°C) means baseline refrigeration load during non-extreme days is materially lower.

4.8.3 Summer (June–August)

Summer is the most operationally critical season for all three product categories. Here, Bragança holds the clearest advantage: 86 fewer days above 25°C, 69 fewer above 30°C, and 29 fewer above 35°C over the 8-year period — averaging 11 fewer high-temperature days per summer season. At ambient temperatures above 35°C, a pharmaceutical cold room maintaining 2–8°C faces a 27–33°C differential across its envelope; each additional degree of ambient temperature above 35°C increases compressor duty cycle and energy draw non-linearly. Bragança's lower Summer positive deviation outlier (maximum ≈10.5°C vs Viseu's ≈12.3°C above monthly mean max) further means that unplanned peak cooling events are less severe. Bragança's Summer Stability Index (0.238 vs 0.216) also confirms more predictable day-to-day temperature behaviour, enabling more reliable energy planning and reducing the likelihood of simultaneous peak demand across multiple cold rooms.

4.8.4 Autumn (September–November)

Viseu records 2 IQR-flagged Autumn extreme events reaching approximately 27–28°C mean daily temperature — an unusual occurrence in a season where ambient cooling would typically allow facilities to reduce refrigeration intensity. No such events are recorded for Bragança. Viseu's Autumn mean (15.30°C vs Bragança's 13.61°C) also means a warmer ambient baseline persists longer into the season, delaying the point at which passive heat dissipation reduces compressor dependency. For luxury chocolate and specialty food products, the Autumn extreme events at Viseu represent a specific risk: inventory brought in from summer at elevated ambient exposure may be stress-tested again before winter storage protocols engage.

4.8.5 Annual Cumulative Assessment

Across the full year, Viseu accumulates 184 additional high-temperature days at the three tracked thresholds (25°C, 30°C, 35°C), averaging 23 extra high-temperature days per year. For a continuous cold-chain operation, this additional load translates to meaningfully higher annual energy consumption, accelerated compressor wear cycles, and reduced system headroom for equipment failure events. The product mix considered here —

pharmaceuticals with regulatory temperature traceability requirements, specialty food with quality-grade sensitivity, and luxury goods with zero tolerance for heat damage — amplifies the cost of each exceedance event relative to a generic ambient warehouse. On balance, Bragança's thermal profile reduces both the frequency and severity of the conditions most likely to generate operational cost overruns, product losses, or compliance failures.

4.9 Robustness Checks

- The IQR-based extreme event detection was validated by profiling flagged days against seasonal meteorological baselines, confirming that all identified events exhibit the expected hot-and-dry signature (elevated radiation, reduced humidity, expanded DTR) rather than data artefacts.
- The Stability Index is constructed from three independently meaningful components (within-season SD, DTR, inter-annual CoV), each capturing a distinct aspect of thermal variability. Convergent agreement across all three components reinforces the directional findings.
- Pearson correlations were computed over the full 2,922-day dataset per city, providing adequate statistical power for the effect sizes observed.
- Results are consistent across the 2018–2025 window with no single year disproportionately influencing aggregate statistics, supporting the representativeness of the findings for long-run operational planning.

Chapter 5

Limitations and Assumptions

5.1 Assumptions

- Reanalysis data from Open-Meteo is assumed to be an accurate representation of local climate conditions at both sites. No station-level ground-truth validation was performed.
- The seasonal assignment (Winter = Dec–Feb, etc.) is assumed to capture the operationally relevant thermal regimes. Alternative seasonal boundaries would affect the Stability Index but are unlikely to change the directional findings.
- Equal weighting of the three Stability Index components is assumed. A client with specific priorities (e.g., greater weight on DTR) could adjust these weights.
- The analysis assumes stationarity over the 2018–2025 period. Climate trends within this window are not modelled.

5.2 Limitations

- The dataset covers 8 years (2018–2025). While sufficient for seasonal summary statistics, it may not capture rare multi-decade climate extremes.
- Only two candidate cities are compared. A broader site-selection analysis might include additional locations.
- The analysis focuses exclusively on temperature-related variables. Other operationally relevant factors — infrastructure, labour costs, transport access — are outside scope.

- The partial year 2026 was excluded. Including it with appropriate seasonal adjustment could provide marginally more recent data but was deemed not worth the introduced bias.
- Wind speed and gusts showed near-zero correlation with temperature targets, suggesting limited predictive value for thermal load at these sites; however, wind remains operationally relevant for building envelope design and was retained for completeness.

Chapter 6

Provisional Recommendation

Directional recommendation: Bragança is the provisionally preferred location for the cold-storage logistics facility.

This recommendation is based on the quantitative comparative analysis presented in Chapter 4 and should be understood as directional at this stage. Formal statistical validation — including hypothesis testing on the observed mean differences and distributional comparisons — is deferred to Milestone 3. The current evidence is sufficient to indicate a consistent directional advantage for Bragança, but the magnitude of several differences (particularly in Summer mean temperature, 0.35°C, and Summer DTR, 0.08°C) is small enough that significance testing will be required to confirm they are not attributable to sampling variation.

The provisional recommendation rests on the following quantitative evidence:

- **Lower mean temperatures in all four seasons** — Winter gap of 2.34°C and Autumn gap of 1.69°C are large relative to within-season SD (≈2.7–4.7°C); Summer and Spring gaps are smaller (0.35°C and 1.63°C respectively) and warrant confirmation.
- **Substantially fewer extreme heat days** — Bragança records 45.3% fewer days ≥35°C (35 vs 64) and 11% fewer days ≥25°C (700 vs 786) over 8 years; these are count differences on a full-population basis and not subject to sampling error.
- **Higher Summer Stability Index** — Bragança scores 0.238 vs Viseu's 0.216, a gap of 0.022 index points. The practical significance of this gap will be assessed in Milestone 3.
- **Lower and less severe Summer thermal stress spikes** — maximum positive deviation outliers of ≈10.5°C vs Viseu's ≈12.3°C above monthly mean maximum.

Viseu's advantages — Winter Stability Index (+0.114 over Bragança), Autumn Stability Index (+0.040), and one fewer IQR-defined extreme event — do not alter the directional finding given the operational weight of Summer and threshold-exceedance metrics for the mixed product profile considered.

Bibliography